IIT ROORKEE

NPTEL ONLINE
CERTIFICATION COURSE

χ^2 Test of Independence - I

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DEPARTMENT OF MANAGEMENT STUDIES

Agenda

- To understand χ^2 Test of Independence

$\bar{x} \to \mu$

1 sample Z-test

1 sample Z proportion test

$\bar{x}_1$ $\bar{x}_2$

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

two sample Z, t

2 Sample Z-proportion test

$H_0: \mu_1 = \mu_2 = \mu_3$

$H_1: \mu_1 \neq \mu_2 \neq \mu_3$

ANOVA

Chi-squared test

χ^2 Test of Independence

- It is used to analyze the frequencies of two variables with multiple categories to determine whether the two variables are independent.
- Qualitative Variables
- Nominal Data

χ^2 Test of Independence: Investment Example

- In which region of the country do you reside?
  A. Northeast B. Midwest C. South D. West
- Which type of financial investment are you most likely to make today?
  E. Stocks F. Bonds G. Treasury bills

Contingency Table

		Type of financial Investment			
		E	F	G	
Geographic Region	A			O_{13}	n_A
	B				n_B
	C				n_C
	D				n_D
		n_E	n_F	n_G	N

χ^2 Test of Independence: Investment Example

If A and F are independent,
$P(A \cap F) = P(A) \cdot P(F)$

$$P(A) = \frac{n_A}{N} \qquad P(F) = \frac{n_F}{N}$$

$$P(A \cap F) = \frac{n_A}{N} \cdot \frac{n_F}{N}$$

$$e_{AF} = N \cdot P(A \cap F)$$

$$= \cancel{N}\left(\frac{n_A}{\cancel{N}} \cdot \frac{n_F}{N}\right)$$

$$= \frac{n_A \cdot n_F}{N}$$

Contingency Table

		Type of Financial Investment			
		E	F	G	
Geographic Region	A		e_{12}		n_A
	B				n_B
	C				n_C
	D				n_D
		n_E	n_F	n_G	N

χ^2 Test of Independence: Formulas

Expected Frequencies

$$e_{ij} = \frac{(n_i)(n_j)}{N}$$

where: i = the row

j = the column

n_i = the total of row i

n_j = the total of column j

N = the total of all frequencies

χ^2 Test of Independence: Formulas

Calculated χ^2
(Observed χ^2)

$$\chi^2 = \sum\sum \frac{(f_o - f_e)^2}{f_e}$$

where: df = (r - 1)(c - 1)
r = the number of rows
c = the number of columns

Example for Independence

χ^2 Test of Independence

H_o : Type of gasoline is independent of income

H_a : Type of gasoline is not independent of income

χ^2 Test of Independence

Type of Gasoline

r = 4

c = 3

Income	Regular	Premium	Extra Premium
Less than $30,000			
$30,000 to $49,999			
$50,000 to $99,000			
At least $100,000			

χ^2 Test of Independence: Gasoline Preference Versus Income Category

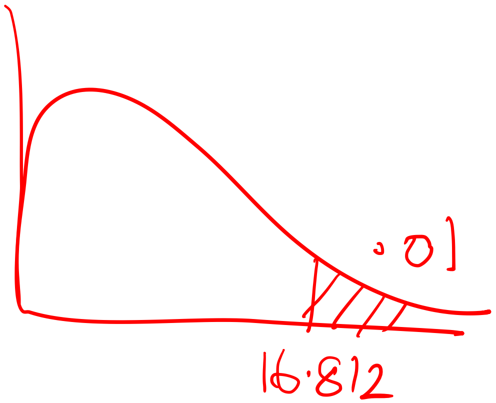
$$\alpha = .01$$

$$df = (r-1)(c-1)$$

$$= (4-1)(3-1)$$

$$= 6$$

$$\chi^2_{.01,6} = 16.812$$



If $\chi^2_{\text{Cal}} > 16.812$, reject Ho.

If $\chi^2_{\text{Cal}} \leq 16.812$, do not reject Ho.

Python code

```
In [5]: import pandas
        import numpy
        from scipy import stats
```

```
In [6]: stats.chi2.ppf(0.99,6)

Out[6]: 16.811893829770927
```

Gasoline Preference Versus Income Category: Observed Frequencies

	Type of Gasoline			
Income	Regular	Premium	Extra Premium	
Less than $30,000	85	16	6	107
$30,000 to $49,999	102	27	13	142
$50,000 to $99,000	36	22	15	73
At least $100,000	15	23	25	63
	238	88	59	385

Gasoline Preference Versus Income Category: Expected Frequencies

$$e_{ij} = \frac{(n_i)(n_j)}{N}$$

$$e_{11} = \frac{(107)(238)}{385} = 66.15$$

$$e_{12} = \frac{(107)(88)}{385} = 24.46$$

$$e_{13} = \frac{(107)(59)}{385} = 16.40$$

Income	Type of Gasoline: Regular	Premium	Extra Premium	
Less than $30,000	(66.15) 85	(24.46) 16	(16.40) 6	107
$30,000 to $49,999	(87.78) 102	(32.46) 27	(21.76) 13	142
$50,000 to $99,000	(45.13) 36	(16.69) 22	(11.19) 15	73
At least $100,000	(38.95) 15	(14.40) 23	(9.65) 25	63
	238	88	59	385

Gasoline Preference Versus Income Category: χ^2 Calculation

$$\chi^2 = \sum\sum \frac{(f_o - f_e)^2}{f_e}$$

$$= \frac{(85-66.15)^2}{66.15} + \frac{(16-24.46)^2}{24.46} + \frac{(6-16.40)^2}{16.40} +$$

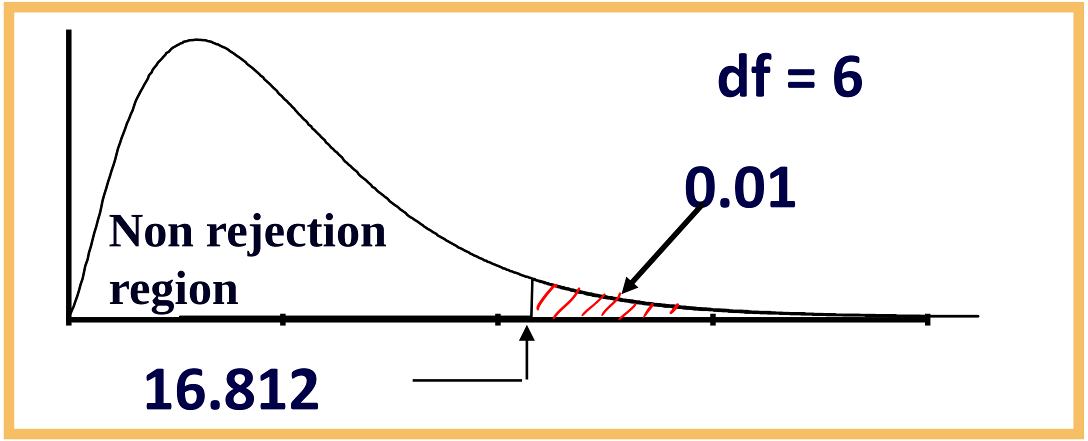
$$\frac{(102-87.78)^2}{87.78} + \frac{(27-32.46)^2}{32.46} + \frac{(13-21.76)^2}{21.76} +$$

$$\frac{(36-45.13)^2}{45.13} + \frac{(22-16.69)^2}{16.69} + \frac{(15-11.19)^2}{11.19} +$$

$$\frac{(15-38.95)^2}{38.95} + \frac{(23-14.40)^2}{14.40} + \frac{(25-9.65)^2}{9.65}$$

$$= 70.75$$

Gasoline Preference Versus Income Category: Conclusion



$$\chi^2_{Cal} = 70.7\cancel{8}5 > 16.812, \text{ reject H}_o.$$

Contingency Tables

Contingency Tables

- Useful in situations involving multiple population proportions
- Used to classify sample observations according to two or more characteristics
- Also called a cross-classification table.

Contingency Table Example

Hand Preference vs. Gender

Dominant Hand: Left vs. Right

Gender: Male vs. Female

- 2 categories for each variable, so the table is called a 2 x 2 table
- Suppose we examine a sample of 300 college students

Contingency Table Example

Sample results organized in a contingency table:

sample size = n = 300:

120 Females, 12 were left handed

180 Males, 24 were left handed

Hand Preference	Gender		
	Female	Male	
Left	12	24	36
Right	108	156	264
	120	180	300

Contingency Table Example

H_0: $\pi_1 = \pi_2$ (Proportion of females who are left handed is equal to the proportion of males who are left handed)

H_1: $\pi_1 \neq \pi_2$ (The two proportions are not the same Hand preference is **not** independent of gender)

- If H_0 is true, then the proportion of left-handed females should be the same as the proportion of left-handed males.
- The two proportions above should be the same as the proportion of left-handed people overall.

The Chi-Square Test Statistic

The Chi-square test statistic is:

$$\chi^2 = \sum_{\text{all cells}} \frac{(f_o - f_e)^2}{f_e}$$

where:

f_o = observed frequency in a particular cell

f_e = expected frequency in a particular cell if H_0 is true

χ^2 for the 2 x 2 case has 1 degree of freedom

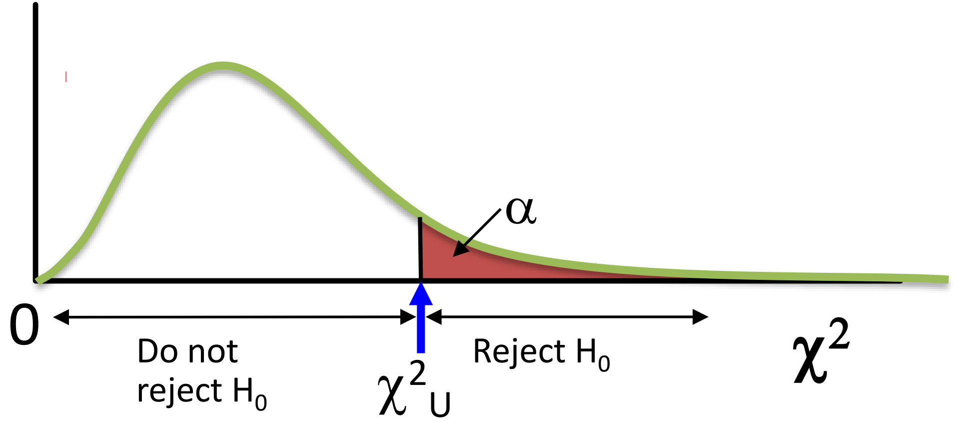
Assumed: each cell in the contingency table has expected frequency of at least 5

The Chi-Square Test Statistic

The χ^2 test statistic approximately follows a chi-square distribution with one degree of freedom

Decision Rule:

If $\chi^2 > \chi^2_U$, reject H_0, otherwise, do not reject H_0



Observed vs. Expected Frequencies

Hand Preference	Gender		
	Female	Male	
Left	Observed = 12 Expected = 14.4 $\frac{36 \times 120}{300}$	Observed = 24 Expected = 21.6 $\frac{36 \times 180}{300}$	36
Right	Observed = 108 Expected = 105.6 $\frac{264 \times 120}{300}$	Observed = 156 Expected = 158.4 $\frac{264 \times 180}{300}$	264
	120	180	300

The Chi-Square Test Statistic

Hand Preference	Gender		
	Female	Male	
Left	Observed = 12 Expected = 14.4	Observed = 24 Expected = 21.6	36
Right	Observed = 108 Expected = 105.6	Observed = 156 Expected = 158.4	264
	120	180	300

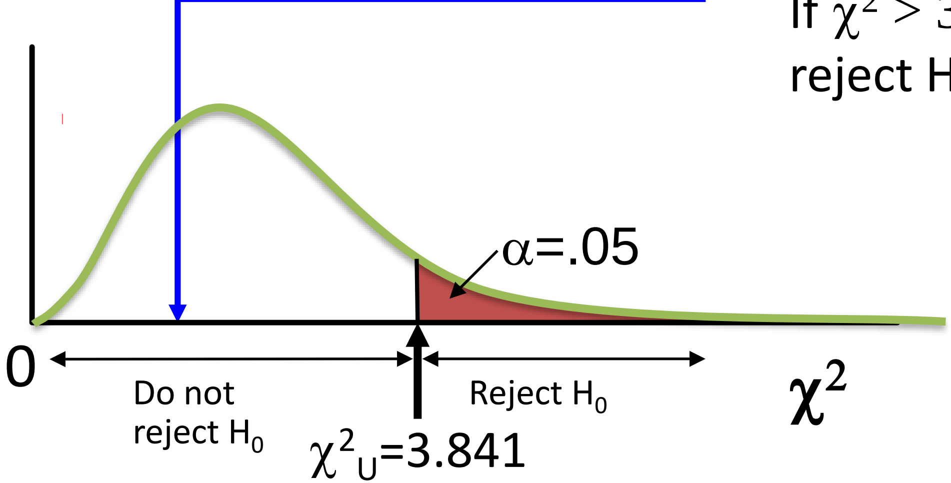
The test statistic is:

$$\chi^2 = \sum_{\text{all cells}} \frac{(f_o - f_e)^2}{f_e}$$

$$= \frac{(12-14.4)^2}{14.4} + \frac{(108-105.6)^2}{105.6} + \frac{(24-21.6)^2}{21.6} + \frac{(156-158.4)^2}{158.4} = \boxed{0.7576}$$

The Chi-Square Test Statistic

The test statistic is $\chi^2 = \boxed{0.7576,}$ χ^2_U with 1 d.f. = 3.841



Decision Rule:

If $\chi^2 > 3.841$, reject H_0, otherwise, do not reject H_0

Here,

$\chi^2 = 0..7576 < \chi^2_U = 3.841$,

so you do not reject H_0 and conclude that there is insufficient evidence that the two proportions are different.

χ^2 Test for The Differences Among More Than Two Proportions

- Extend the χ^2 test to the case with more than two independent populations:

H_0: $\pi_1 = \pi_2 = \ldots = \pi_c$

H_1: Not all of the π_j are equal ($j = 1, 2, \ldots, c$)

The Chi-Square Test Statistic

The Chi-square test statistic is:

$$\chi^2 = \sum_{\text{all cells}} \frac{(f_o - f_e)^2}{f_e}$$

where:

- f_o = observed frequency in a particular cell of the 2 x c table
- f_e = expected frequency in a particular cell if H_0 is true
- χ^2 for the 2 x c case has (2-1)(c-1) = c - 1 degrees of freedom

Assumed: each cell in the contingency table has expected frequency of at least 5

χ^2 Test with More Than Two Proportions: Example

The sharing of patient records is a controversial issue in health care. A survey of 500 respondents asked whether they objected to their records being shared by insurance companies, by pharmacies, and by medical researchers. The results are summarized on the following table:

χ2 Test with More Than Two Proportions: Example

Object to Record Sharing	Organization		
	Insurance Companies π_1	Pharmacies π_2	Medical Researchers π_3
Yes	410	295	335
No	90	205	165

χ2 Test with More Than Two Proportions: Example

Object to Record Sharing	Organization			
	Insurance Companies	Pharmacies	Medical Researchers	Row Sum
Yes	410 $\frac{1040 \times 500}{1500}$	295 $\frac{1040 \times 500}{1500}$	335	1040
No	90	205	165	460
Column Sum	500	500	500	1500

χ2 Test with More Than Two Proportions: Example

The overall proportion is:

$$\bar{p} = \frac{X_1 + X_2 + \ldots + X_c}{n_1 + n_2 + \ldots + n_c} = \frac{410 + 295 + 335}{500 + 500 + 500} = 0.6933$$

Object to Record Sharing	Organization		
	Insurance Companies	Pharmacies	Medical Researchers
Yes	$f_o = 410$ $f_e = 346.667$	$f_o = 295$ $f_e = 346.667$	$f_o = 335$ $f_e = 346.667$
No	$f_o = 90$ $f_e = 153.333$	$f_o = 205$ $f_e = 153.333$	$f_o = 165$ $f_e = 153.333$

$\chi 2$ Test with More Than Two Proportions: Example

Object to Record Sharing	Organization		
	Insurance Companies	Pharmacies	Medical Researchers
Yes	$\frac{(f_o - f_e)^2}{f_e} = 11.571$	$\frac{(f_o - f_e)^2}{f_e} = 7.700$	$\frac{(f_o - f_e)^2}{f_e} = 0.3926$
No	$\frac{(f_o - f_e)^2}{f_e} = 26.159$	$\frac{(f_o - f_e)^2}{f_e} = 17.409$	$\frac{(f_o - f_e)^2}{f_e} = 0.888$

The Chi-square test statistic is:

$$\chi^2 = \sum_{\text{all cells}} \frac{(f_o - f_e)^2}{f_e} = 64.1196$$

χ^2 Test with More Than Two Proportions: Example

H_0: $\pi_1 = \pi_2 = \pi_3$

H_1: Not all of the π_j are equal (j = 1, 2, 3)

Decision Rule:
If $\chi^2 > \chi^2_U$, reject H_0, otherwise, do not reject H_0

χ^2_U = 5.991 is from the chi-square distribution with 2 degrees of freedom.

$(2-1)(3-1) = 1 \times 2 = 2$

Conclusion: Since 64.1196 > 5.991, you reject H_0 and you conclude that at least one proportion of respondents who object to their records being shared is different across the three organizations

Thank You

χ^2 Test of Independence - II

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DEPARTMENT OF MANAGEMENT STUDIES

Agenda

- Using python to test the independence of variables
- Understanding goodness of fit test for Poisson

Example

- Record of 50 students studying in ABN School is taken at random, the first 10 entries are like this:

res_num	aa	pe	sm	ae	r	g	c
1	99	19	1	2	0	0	1
2	46	12	0	0	0	0	0
3	57	15	1	1	0	0	0
4	94	18	2	2	1	1	1
5	82	13	2	1	1	1	1
6	59	12	0	0	2	0	0
7	61	12	1	2	0	0	0
8	29	9	0	0	1	1	0
9	36	13	1	1	0	0	0
10	91	16	2	2	1	1	0

Example

Here :

- res_num = registration no.
- aa= academic ability
- pe = parent education
- sm = student motivation
- r = religion
- g = gender

Python code

```
In [1]: import pandas as pd
        import numpy as np

In [2]: acad = pd.read_csv('AcademicAbilityData.csv')

In [3]: acad
```

Out[3]:

	res_num	aa	pe	sm	ae	r	g	c
0	1	99	19	1	2	0	0	1
1	2	46	12	0	0	0	0	0
2	3	57	15	1	1	0	0	0
3	4	94	18	2	2	1	1	1
4	5	82	13	2	1	1	1	1
5	6	59	12	0	0	2	0	0
6	7	61	12	1	2	0	0	0
7	8	29	9	0	0	1	1	0
8	9	36	13	1	1	0	0	0
9	10	91	16	2	2	1	1	0
10	11	55	10	0	0	1	0	0
11	12	58	11	0	1	0	0	0

Hypothesis

- Test the hypothesis that “gender and student motivation” are independent

Python code

```
In [19]: #Cross table between gender and student's motivation
         obs =pd.pivot_table(acad[['g','sm']],index = 'g',columns='sm',aggfunc=len)
         obs
```

Out[19]:

sm	0	1	2
g			
0	10	13	6
1	4	9	8

Observed values

Gender	Student motivation			
	0 (Disagree)	1 (Not decided)	2 (Agree)	Row Sum
0 (Male)	10	13	6	29
1(Female)	4	9	8	21
Column Sum	14	22	14	50

Expected frequency (contingency table)

Gender	Student motivation		
	0	1	2
0	29*14/50= 8.12	12.76	8.12
1	5.88	9.24	5.88

Frequency Table

Gender	Student motivation		
	0	1	2
0	$f_o = 10$ $f_e = 8.12$	$f_o = 13$ $f_e = 12.76$	$f_o = 6$ $f_e = 8.12$
1	$f_o = 4$ $f_e = 5.88$	$f_o = 9$ $f_e = 9.24$	$f_o = 8$ $f_e = 5.88$

Chi sq. calculation

$$\chi^2 = \sum\sum \frac{\left(f_o - f_e\right)^2}{f_e}$$

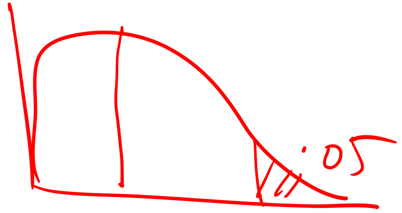
$= 0.435+ 0.005+0.554+0.601+0.006+0.764$

$= 2.365$

Python code

```
In [11]: ## Perform chi2 test to check independence
         from scipy.stats import chi2_contingency

In [14]: chi2, p,dof,tbl= chi2_contingency(obs)

In [15]: chi2

Out[15]: 2.3649585225939904

In [16]: p

Out[16]: 0.3065178579178871

In [17]: dof

Out[17]: 2
```

$\alpha = 5\% = 0.05$



$(2-1)(3-1) = 2$

Python code

```
In [11]: ## Perform chi2 test to check independence
         from scipy.stats import chi2_contingency

In [14]: chi2, p,dof,tbl= chi2_contingency(obs)

In [15]: chi2

Out[15]: 2.3649585225939904

In [16]: p

Out[16]: 0.3065178579178871

In [17]: dof

Out[17]: 2
```

Degrees of freedom = (2-1)*(3-1)

Python code

```
In [12]:  tbl

Out[12]: array([[ 8.12, 12.76,  8.12],
                [ 5.88,  9.24,  5.88]])
```

Contingency table

χ^2 Goodness of Fit Test

χ^2 Goodness-of-Fit Test

- The χ^2 goodness-of-fit test compares *expected* (theoretical) *frequencies* of categories from a population distribution to the *observed* (actual) *frequencies* from a distribution to determine whether there is a difference between what was expected and what was observed

χ^2 Goodness-of-Fit Test

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

$$\text{df} = k - 1 - \text{p}$$

$where:$ $f_o =$ frequency of observed values

$f_e =$ frequency of expected values

$k =$ number of categories

p = number of parameters estimated from the sample data

Goodness of Fit Test: Poisson Distribution

1. Set up the null and alternative hypotheses.

   H_0: Population has a Poisson probability distribution

   H_a: Population does not have a Poisson distribution

2. Select a random sample and

   - Record the observed frequency f_i for each value of the Poisson random variable.
   - Compute the mean number of occurrences μ.

3. Compute the expected frequency of occurrences e_i for each value of the Poisson random variable.

Goodness of Fit Test: Poisson Distribution

4. Compute the value of the test statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(f_i - e_i)^2}{e_i}$$

where:

f_i = observed frequency for category i

e_i = expected frequency for category i

k = number of categories

Goodness of Fit Test: Poisson Distribution

5. Rejection rule:

p-value approach: Reject H_0 if *p*-value $\leq \alpha$

Critical value approach: Reject H_0 if $\boxed{\chi^2 \geq \chi^2_\alpha}$

where α is the significance level and there are $\underline{k - 2}$ degrees of freedom

$K-1-P$

Goodness of Fit Test: Poisson Distribution

- Example: Parking Garage

  In studying the need for an additional entrance to a city parking garage, a consultant has recommended an analysis, that approach is applicable only in situations where the number of cars entering during a specified time period follows a Poisson distribution.

Goodness of Fit Test: Poisson Distribution

A random sample of 100 one- minute time intervals resulted in the customer arrivals listed below. A statistical test must be conducted to see if the assumption of a Poisson distribution is reasonable.

# Arrivals	0	1	2	3	4	5	6	7	8	9	10	11	12
Frequency	0	1	4	10	14	20	12	12	9	8	6	3	1

Goodness of Fit Test: Poisson Distribution

- Hypotheses

H_0: Number of cars entering the garage during a one-minute interval is Poisson distributed

H_a: Number of cars entering the garage during a one-minute interval is <u>not</u> Poisson distributed

Python Code

In [1]:
```
import scipy
from scipy.stats import chi2
from scipy.stats import poisson
```

In [2]:
```
import pandas as pd
import numpy as np
```

In [3]:
```
data = pd.read_excel('P_distribution.xlsx')
data
```

Out[3]:

	Arrivals	Frequency
0	0	0
1	1	1
2	2	4
3	3	10
4	4	14
5	5	20
6	6	12
7	7	12
8	8	9
9	9	8
10	10	6
11	11	3
12	12	1

Goodness of Fit Test: Poisson Distribution

- Estimate of Poisson Probability Function

Total Arrivals = 0(0) + 1(1) + 2(4) + . . . + 12(1) = 600

Estimate of μ = 600/100 = 6

Total Time Periods = 100

Hence,

$$= \frac{e^{-u} u^{x}}{x!}$$

$$f(x) = \frac{6^x e^{-6}}{x!}$$

$$u = \frac{\Sigma f n}{\Sigma f} = \frac{600}{100} = 6$$

Goodness of Fit Test: Poisson Distribution

- Expected Frequencies

x	$f(x)$	$nf(x)$	x	$f(x)$	$nf(x)$
0	.0025	.25	7	.1377	13.77
1	.0149	1.49	8	.1033	10.33
2	.0446	4.46	9	.0688	6.88
3	.0892	8.92	10	.0413	4.13
4	.1339	13.39	11	.0225	2.25
5	.1606	16.06	12+	.0201	2.01
6	.1606	16.06	Total	1.0000	100.00

Python code

```
In [4]: Observed_Freq = data['Frequency']

In [5]: total_arrival = 600
        total_time_period = 100
        mu = total_arrival/total_time_period

In [6]: Expected_Freq = []
        for i in range(len(Observed_Freq)):
            E_Freq = 100*poisson.pmf(i, mu)
            Expected_Freq.append(E_Freq)

In [7]: Expected_Freq

Out[7]: [0.24787521766663584,
         1.4872513059998145,
         4.461753917999444,
         8.923507835998894,
         13.385261753998332,
         16.062314104797995,
         16.06231410479801,
         13.767697804112569,
         10.32577335308442,
         6.883848902056284,
         4.130309341233764,
         2.2528960043093247,
         1.1264480021546681]
```

Python code

```
In [8]: Expected_Freq_round_off = [round(elem, 2) for elem in Expected_Freq]
        Expected_Freq_round_off

Out[8]: [0.25,
         1.49,
         4.46,
         8.92,
         13.39,
         16.06,
         16.06,
         13.77,
         10.33,
         6.88,
         4.13,
         2.25,
         1.13]
```

```
In [9]: df = pd.DataFrame(list(zip(Observed_Freq, Expected_Freq_round_off )), columns = ['Observed Frequency', 'Expected Frequency'])
        df
```

Out[9]:

	Observed Frequency	Expected Frequency
0	0	0.25
1	1	1.49
2	4	4.46
3	10	8.92
4	14	13.39
5	20	16.06
6	12	16.06
7	12	13.77
8	9	10.33
9	8	6.88
10	6	4.13
11	3	2.25
12	1	1.13

Goodness of Fit Test: Poisson Distribution

- Observed and Expected Frequencies

i	f_i	e_i	$f_i - e_i$
0 or 1 or 2	5	6.20	-1.20
3	10	8.92	1.08
4	14	13.39	0.61
5	20	16.06	3.94
6	12	16.06	-4.06
7	12	13.77	-1.77
8	9	10.33	-1.33
9	8	6.88	1.12
10 or more	10	8.39	1.61

Python code

```
In [10]: obs_freq = [5, 10, 14, 20, 12, 12, 9, 8, 10]
         expected_freq = [6.20, 8.92, 13.39, 16.06, 16.06, 13.77, 10.33, 6.88, 8.39]

In [11]: scipy.stats.chisquare(obs_freq, expected_freq)

Out[11]: Power_divergenceResult(statistic=3.2738182931105193, pvalue=0.916017731732134)
```

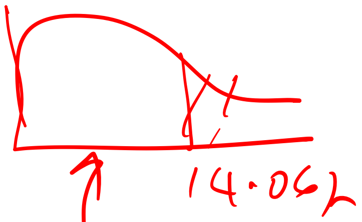
Goodness of Fit Test: Poisson Distribution

- Rejection Rule

With $\alpha = .05$ and $k - p - 1 = 9 - 1 - 1 = 7$ d.f.
(where k = number of categories and p = number of population parameters estimated),

$$\chi^2_{.05} = 14.067$$

Reject H_0 if p-value $\leq .05$ or $\chi^2 \geq 14.067$.



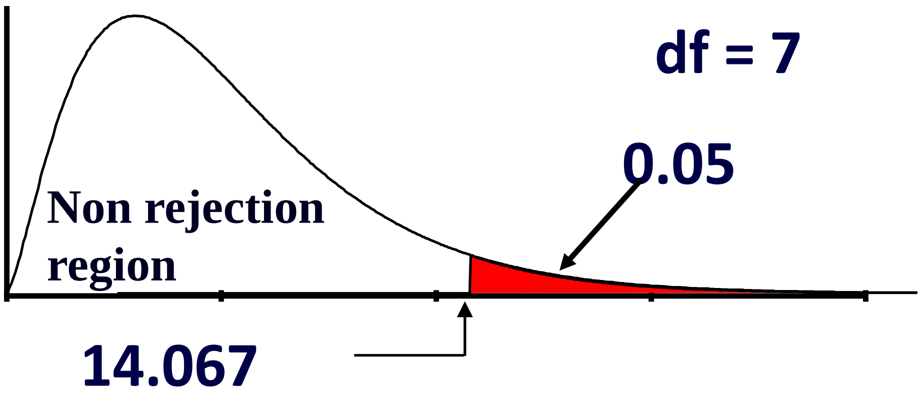
- Test Statistic

$$\chi^2 = \frac{(-1.20)^2}{6.20} + \frac{(1.08)^2}{8.92} + \ . \ . \ . \ + \frac{(1.61)^2}{8.39} = 3.268$$

Python code

```
In [4]: from scipy.stats import chi2
        chi2.ppf(0.95,7)

Out[4]: 14.067140449340167
```

Goodness of Fit Test: Poisson Distribution



$$\chi^2_{Cal} = 3.268 < 14.067, \text{ do not reject H}_o.$$

Thank You

χ^2 Goodness of Fit Test

Dr. A. Ramesh
DEPARTMENT OF MANAGEMENT STUDIES

Agenda

- Python demo for testing GOF for Poisson distribution
- Understanding goodness of fit test for:
  - Uniform
  - Normal
- Python demo for testing GOF for uniform and normal distribution

Goodness of fit for Uniform Distribution

- Milk Sales Data

Month	Litres
January	1,610
February	1,585
March	1,649
April	1,590
May	1,540
June	1,397
July	1,410
August	1,350
September	1,495
October	1,564
November	1,602
December	1,655
	18,447

Hypotheses and Decision Rules

H_o: The monthly milk figures for milk sales are uniformly distributed

H_a: The monthly milk figures for milk sales are not uniformly distributed

$$\alpha = .01$$
$$df = k - 1 - p$$
$$= 12 - 1 - 0$$
$$= 11$$
$$\chi^2_{.01,11} = 24.725$$

If $\chi^2_{Cal} > 24.725$, reject H_o.

If $\chi^2_{Cal} \le 24.725$, do not reject H_o.

Python code

```
In [1]: from scipy.stats import chi2

In [2]: import pandas as pd
        import numpy as np

In [3]: chi2.ppf(0.99,11)

Out[3]: 24.724970311318277
```

Calculations

Month	f_o	f_e	$(f_o - f_e)^2/f_e$
January	1,610	1,537.25	3.44
February	1,585	1,537.25	1.48
March	1,649	1,537.25	8.12
April	1,590	1,537.25	1.81
May	1,540	1,537.25	0.00
June	1,397	1,537.25	12.80
July	1,410	1,537.25	10.53
August	1,350	1,537.25	22.81
September	1,495	1,537.25	1.16
October	1,564	1,537.25	0.47
November	1,602	1,537.25	2.73
December	1,655	1,537.25	9.02
	18,447	18,447.00	74.38

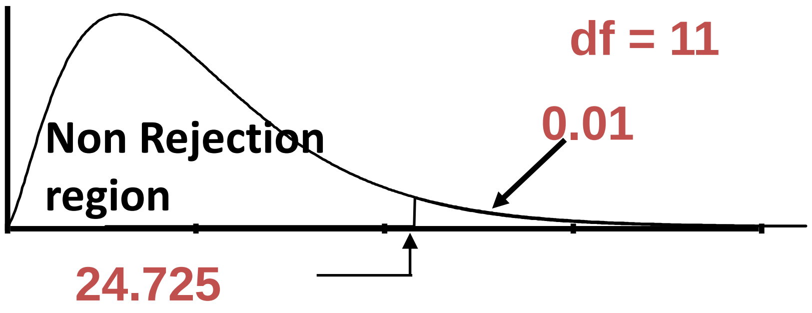
$$f_e = \frac{18447}{12} = 1537.25$$

$$\chi^2_{Cal} = 74.37$$

Python code

```
In [6]: x =[1610,1585,1649,1590,1540,1397,1410,1350,1495,1564,1602,1655]

In [7]: np.mean(x)

Out[7]: 1537.25

In [8]: exp_f=[1537.25,1537.25,1537.25,1537.25,1537.25,1537.25,1537.25,1537.25,1537.25,1537.25,1537.25,1537.25]

In [9]: from scipy.stats import chisquare
        chisquare(x,exp_f)

Out[9]: Power_divergenceResult(statistic=74.37583346885673, pvalue=1.785452527830340e-11)
```

Conclusion



$$\chi^2_{Cal} = 74.37 > 24.725, \text{ reject } H_o.$$

Goodness of Fit Test: Normal Distribution

1. Set up the null and alternative hypotheses.
2. Select a random sample and
   a. Compute the mean and standard deviation.
   b. Define intervals of values so that the expected frequency is at least 5 for each interval.
   c. For each interval record the observed frequencies
3. Compute the expected frequency, e_i, for each interval.

Goodness of Fit Test: Normal Distribution

4. Compute the value of the test statistic.

$$\chi^2 = \sum_{i=1}^{k} \frac{(f_i - e_i)^2}{e_i}$$

5. Reject H_0 if $\chi^2 \geq \chi^2_\alpha$

   (where α is the significance level and there are k - 3 degrees of freedom)

$k-1-p$

Normal Distribution Goodness of Fit Test

- Example: IQL Computers

IQL Computers manufactures and sells a general purpose microcomputer. As part of a study to evaluate sales personnel, management wants to determine, at $\alpha = 0.05$ significance level, if the annual sales volume (number of units sold by a salesperson) follows a normal probability distribution.

Normal Distribution Goodness of Fit Test

A simple random sample of 30 of the salespeople was taken and their numbers of units sold are below.

33	43	44	45	52	52	56	58	63	64
64	65	66	68	70	72	73	73	74	75
83	84	85	86	91	92	94	98	102	105

(mean = 71, standard deviation = 18.23)

Python code

```
In [12]: A =[33, 43, 44, 45, 52, 52, 56, 58, 63, 64, 64, 65, 66, 68, 70, 72, 73, 73, 74, 75, 83, 84, 85, 86, 91, 92, 94, 98, 102, 105]

In [13]: mean = np.mean(A)
         mean

Out[13]: 71.0

In [14]: std = np.std(A)
         std

Out[14]: 18.226354544998845
```

Normal Distribution Goodness of Fit Test

- Hypotheses

H_0: The population of number of units sold
has a normal distribution with mean 71
and standard deviation 18.23

H_a: The population of number of units sold
does not have a normal distribution with
mean 71 and standard deviation 18.23

Normal Distribution Goodness of Fit Test

- Interval Definition

To satisfy the requirement of an expected frequency of at least 5 in each interval we will divide the normal distribution into 30/5 = 6 equal probability intervals.

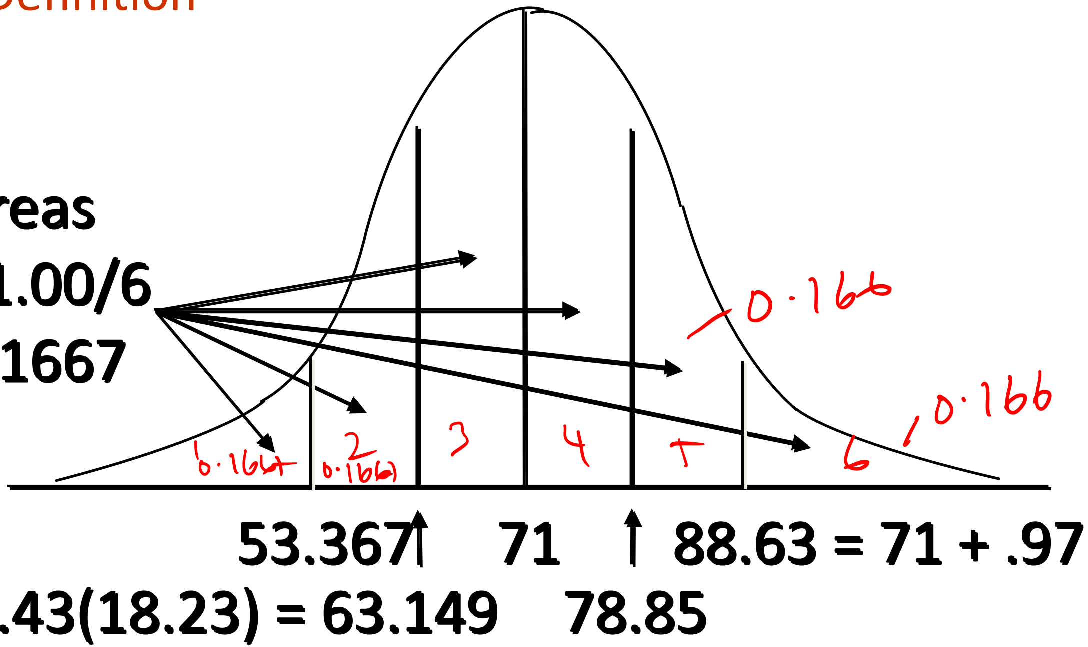
Normal Distribution Goodness of Fit Test

- Interval Definition

Areas

= 1.00/6

= .1667



$\bar{x} - 2\sigma$

Python code

```
In [15]: x = 1/6 #for 6 equal probability intervals.

In [16]: for j in range(1,6):
             Prob_intervals = [scipy.stats.norm.ppf(j*x, mean, std)]
             print ( Prob_intervals)

[53.36743154175236]
[63.14941153083116]
[71.0]
[78.85058846916884]
[88.63256845824763]
```

Normal Distribution Goodness of Fit Test

- Observed and Expected Frequencies

i	f_i	e_i	$f_i - e_i$
Less than 53.02	6	5	1
53.02 to 63.03	3	5	-2
63.03 to 71.00	6	5	1
71.00 to 78.97	5	5	0
78.97 to 88.98	4	5	-1
More than 88.98	6	5	1
Total	30	30	

$\frac{30}{6} = 5$

Python code

```
In [17]: Expected_Freq = [5,5,5,5,5,5] #will divide the normal distribution into 6 intervals at frequency 5 in each

In [18]: Obs_f = [6,3,6,5,4,6]

In [19]: scipy.stats.chisquare(Obs_f, Expected_Freq)

Out[19]: Power_divergenceResult(statistic=1.5999999999999999, pvalue=0.9012493445012737)
```

$\alpha = 5\%$

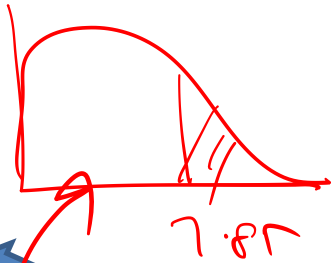
Normal Distribution Goodness of Fit Test

- Rejection Rule

With $\alpha = .05$ and $k - p - 1 = 6 - 2 - 1 = 3$ d.f.

(where k = number of categories and p = number of population parameters estimated), $\chi^2_{.05} = 7.815$

Reject H_0 if p-value $\leq .05$ or $\chi^2 \geq 7.815$.



```
In [5]: chi2.ppf(0.95,3)
Out[5]: 7.8147279032511765
```

- Test Statistic

$$\chi^2 = \frac{(1)^2}{5} + \frac{(-2)^2}{5} + \frac{(1)^2}{5} + \frac{(0)^2}{5} + \frac{(-1)^2}{5} + \frac{(1)^2}{5} = 1.600$$

Thank you

Cluster analysis: Introduction - I

Dr. A. Ramesh
DEPARTMENT OF MANAGEMENT STUDIES

Agenda

- Understanding cluster analysis and its purpose
- Introduction to types of data and how to handle them

Cluster Analysis

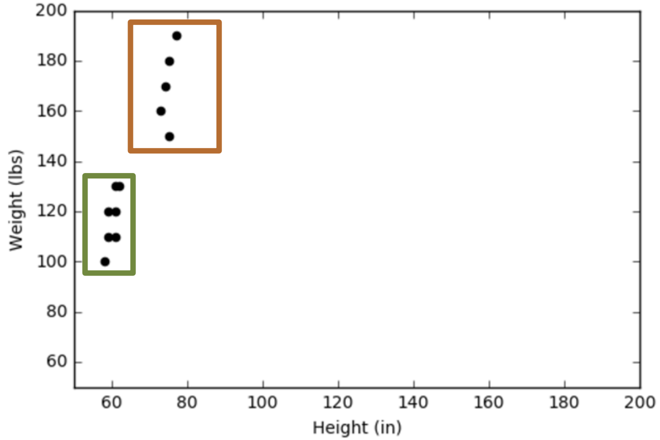
- Cluster analysis is the art of finding groups in data
- In cluster analysis basically, one wants to form groups in such a way that objects in the same group are similar to each other, whereas objects in different groups are as dissimilar as possible

Cluster analysis

- The classification of similar objects into groups is an important human activity, this is part of the learning process
- i.e. A child learns to distinguish between cats and dogs, between tables and chairs, between men and women, by means of continuously improving subconscious classification schemes
- This explains why cluster analysis is often considered as a branch of pattern recognition and artificial intelligence

Example

- Lets illustrate with the help of an example:
- It is a plot of twelve objects, on which two variables were measured. For instance, the weight of an object might be displayed on the vertical axis and its height on the h



Example

- Because this example contains only two variables, we can investigate it by merely looking at the plot
- In this small data set there are clearly two distinct groups of objects
- Such groups are called clusters, and to discover them is the aim of cluster analysis

Cluster and discriminant analysis

- Cluster Analysis is an unsupervised classification technique in the sense that it is applied to a dataset where patterns want to be discovered (i.e. groups of individuals or variables want to be found)
- No prior knowledge is needed for this grouping, and it is sensitive to several decisions that have to be taken (similarity/dissimilarity measures, clustering method,...)

- Discriminant Analysis (DA) is a statistical technique used to build a prediction model that is used to classify objects from a dataset depending on the features observed on them. In this case, the dependent variable is the grouping variable, which identifies to which group and object belongs
- This grouping variable should be known at the beginning, for the function to be built up. Sometimes DA is considered as a Supervised tool, as there is a previous known classification for the elements of the dataset

Cluster analysis and discriminant analysis

- Cluster analysis can be used not only to identify a structure already present in the data, but also to impose a structure on a more or less homogeneous data set that has to be split up in a "fair" way, for instance when dividing a country into telephone areas
- Cluster a from discriminant analysis in that it actually whereas discriminant analysis assigns objects ned in advance

Telephone area code for USA

Types of data and how to handle them

- Let us take an example, there are n objects to be clustered, which may be persons, flowers, words, countries, or anything
- Clustering algorithms typically operate on either of two input structures:
  - The first represents the objects by means of p measurements or attributes, such as height, weight, sex, color, and so on
  - These measurements can be arranged in an n-by-p matrix, where the rows correspond to the objects and the columns to the attributes

Example

Attributes

Objects

	Price	*Quality*	*Time*
Like	*A*	*B*	*B*
Intermediate	*B*	*A*	*A*
Need	*C*	*C*	*C*

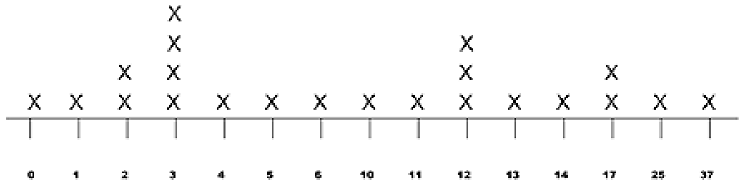
Types of data and how to handle them

- The second structure is a collection of proximities that must be available for all pairs of objects
- These proximities make up an n-by-n table, which is called a one-mode matrix because the row and column entities are the same set of objects
- one shall consider two types of proximities, namely dissimilarities (which measure how far away two objects are from each other) and similarities (which measure how much they resemble each other)

	A	B	C
A	1		
B		1	
C			1

Type of data

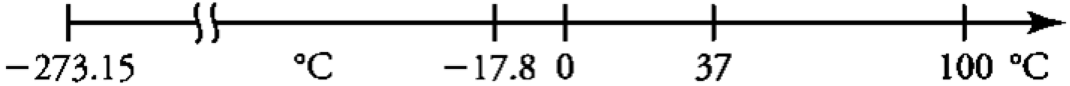
- **Interval-Scaled Variables**
- In this situation the n objects are characterized by p continuous measurements
- These values are positive or negative real numbers, such as height, weight, temperature, age, cost, ..., which follow a linear scale
- For instance, th 1910 was equal in length to that betweer



Time scale in years

Type of data

- Also, it takes the same amount of energy to heat an object of -16.4°C to -12.4°C as to increase it from 35.2°C to 39.2°C
- In general it is required that intervals keep the same importance throughout the scale



Interval-Scaled Variables

- These measurements can be organized in an n-by-p matrix, where the rows correspond to the objects (or cases) and the columns correspond to the variables.
- When the f^{th} measurement of the i^{th} object is denoted by x_{if} (where i = 1,. . . . , n and f = 1

$$
\begin{array}{cc}
 & p \text{ variables} \\
n \text{ objects} & \begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \vdots & & \vdots & & \vdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \vdots & & \vdots & & \vdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}
\end{array}
$$

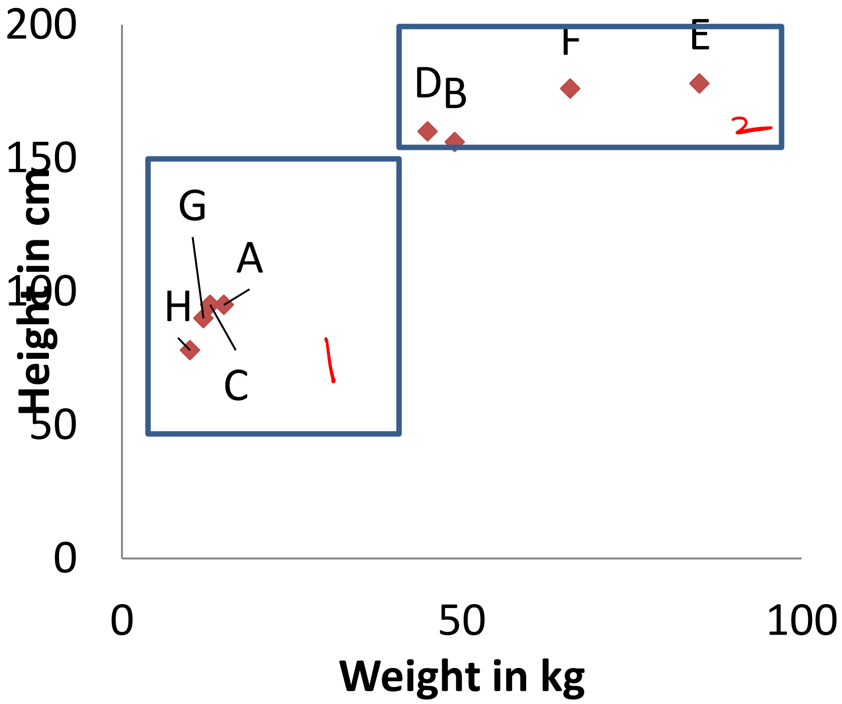
Interval-Scaled Variables

- For example :
- Take eight people, the weight (in kilograms) and the height (in centimetres
- In this situation, n = 8 and p = 2.

Person	Weight(Kg)	Height(cm)
A	15	95
B	49	156
C	13	95
D	45	160
E	85	178
F	66	176
G	12	90
H	10	78

Table :1

Figure 1



Interval-Scaled Variables

- The units on the vertical axis are drawn to the same size as those on the horizontal axis, even though they represent different physical concepts
- The plot contains two obvious clusters, which can in this case be interpreted easily: the one consists of small children and the other of adults
- However, other variables might have led to completely different clustering
- For instance, measuring the concentration of certain natural hormones might have yielded a clear cut partition into different male and female persons

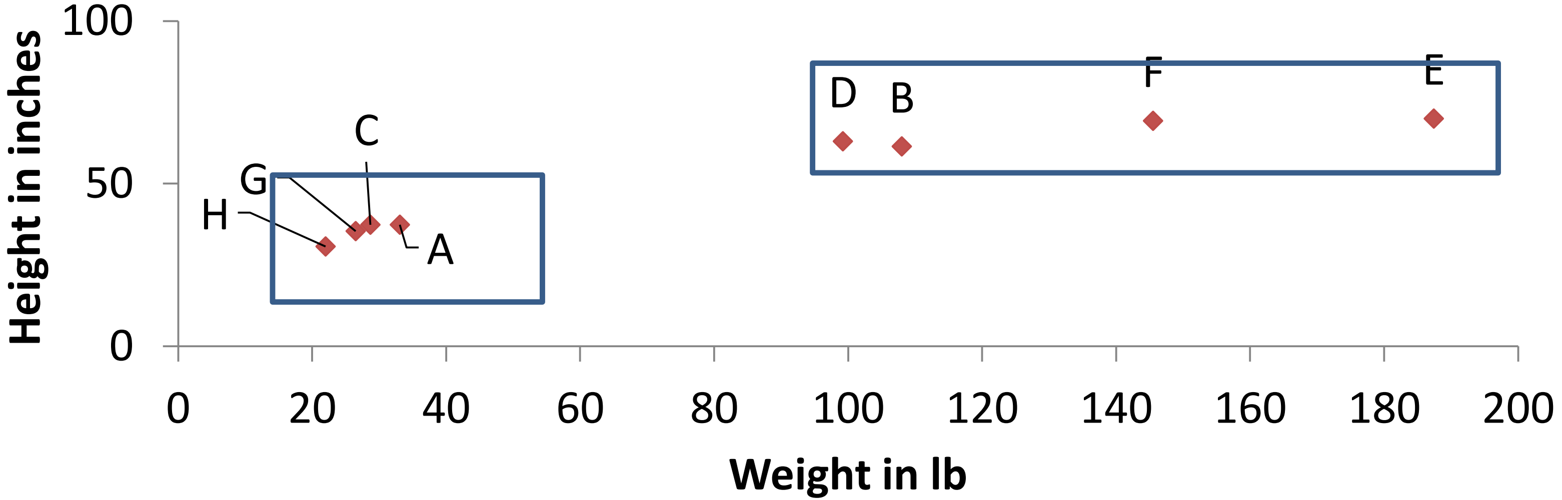
Interval-Scaled Variables

- Let us now consider the effect of changing measurement units.
- If weight and height of the subjects had been expressed in pounds and inches, the results would have looked quite different.
- A pound equals 0.4536 kg and an inch is 2.54 cm
- Therefore, Table 2 contains larger numbers in the column of weights and smaller numbers in the column of heights. Figure 2

Person	Weight(lb)	Height(in)
A	33.1	37.4
B	108	61.4
C	28.7	37.4
D	99.2	63
E	187.4	70
F	145.5	69.3
G	26.5	35.4
H	22	30.7

Table :2

Figure 2



100
50
0
Height in inches
0
20
40
60
80
100
120
140
160
180
200
Weight in lb
A
B
C
D
E
F
G
H

Interpretation

- Although plotting essentially the same data as Figure 1, Figure 2 looks much flatter
- In this figure, the relative importance of the variable “weight” is much larger than in Figure 1
- As a consequence, the two clusters are not as nicely separated as in Figure 1 because in this particular example the height of a person gives a better indication of adulthood than his or her weight. If height had been expressed in feet (1 ft = 30.48 cm), the plot would become flatter still and the variable “weight” would be rather dominant
- In some applications, changing the measurement units may even lead one to see a very different clustering structure

Standardizing the data

- To avoid this dependence on the choice of measurement units, one has the option of standardizing the data
- This converts the original measurements to unitless variables
- First one calculates the m ven by:

$$m_f = \frac{1}{n}(x_{1f} + x_{2f} + \cdots + x_{nf})$$

for each f = 1,. . . , p

Standardizing the data

- Then one computes a measure of the dispersion or “spread” of this fth variable
- Genera''

$$\mathrm{std}_f = \sqrt{\frac{1}{n-1}\{(x_{1f}-m_f)^2 + (x_{2f}-m_f)^2 + \cdots + (x_{nf}-m_f)^2\}}$$

Standardizing the data

- However, this measure is affected very much by the presence of outlying values
- For instance, suppose that one of the x_{if} has been wrongly recorded, so that it is much too large
- In this case std_f will MAD ~~be unduly inflated~~, because x_{if} - m_f is squared
- Hartigan (1975, p. 299) notes that one needs a dispersion measure that is not too sensitive to outliers
- Therefore, we
$$s_f = \frac{1}{n}\{|x_{1f} - m_f| + |x_{2f} - m_f| + \cdots + |x_{nf} - m_f|\}$$
he contribution of … bsolute value $|x_{if} - m_f|$

Standardizing the data

- Let us assume that s_f is nonzero (otherwise variable f is constant over all objects and must be removed)
- Then the standardized measurements are defined by and sometimes called z-scores
- They are unitless because both the numerator and the denominator are expressed in the same uni

$$z_{if} = \frac{x_{if} - m_f}{s_f}$$

- By construction, the z_{if} hav and their mean absolute deviation is equal to1

(0, 1)

Standardizing the data

- When applying standardization, one forgets about the original data and uses the new data matrix in all subsequent computations

$$
\begin{array}{cc}
 & \text{variables} \\
\text{objects} & \begin{bmatrix}
z_{11} & \cdots & z_{1f} & \cdots & z_{1p} \\
\vdots & & \vdots & & \vdots \\
z_{i1} & \cdots & z_{if} & \cdots & z_{ip} \\
\vdots & & \vdots & & \vdots \\
z_{n1} & \cdots & z_{nf} & \cdots & z_{np}
\end{bmatrix}
\end{array}
$$

Detecting outlier

- The advantage of using s_f rather than std_f, in the denominator of z-score formula is that s_f will not be blown up so much in the case of an outlying x_{lf}, and hence the corresponding z_{if} will still be noticeable so the i^{th} object can be recognized as an outlier by the clustering algorithm, which will typically put it in a separate cluster

Standardizing the data

- The preceding description might convey the impression that standardization would be beneficial in all situations.
- However, it is merely an option that may or may not be useful in a given application
- Sometimes the variables have an absolute meaning, and should not be standardized
- For instance, it may happen that several variables are expressed in the same units, so they should not be divided by different s_f
- Often standardization dampens a clustering structure by reducing the large effects because the variables with a big contribution are divided by a large s_f

Thank you

IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Cluster analysis: Part - II

Dr. A. Ramesh

DEPARTMENT OF MANAGEMENT STUDIES

Agenda

- Explain effect of standardization(with help of an example)
- Different types of distances computation between the objects

Example

- Lets take four persons A, B,C, D with following age and height:

Person	Age (yr)	Height (cm)
A	35	190
B	40	190
C	35	160
D	40	160

TABLE: 1

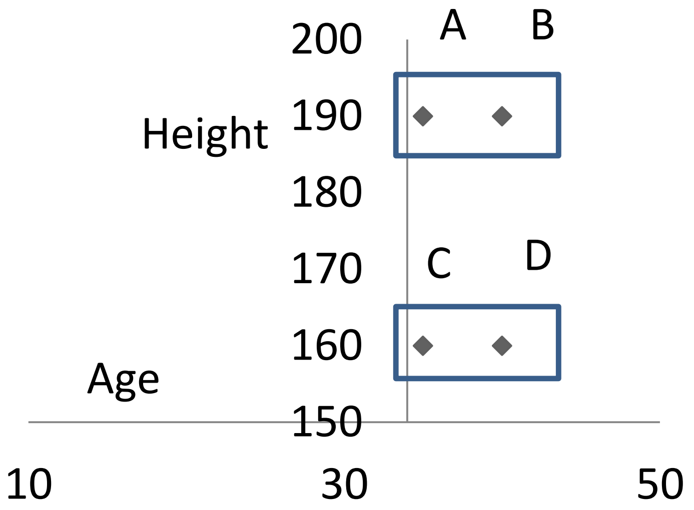


FIGURE: 1

Finding Groups in Data: An Introduction to Cluster Analysis
Author(s): Leonard Kaufman, Peter J. Rousseeuw
March 1990, John Wiley & Sons, Inc.

Example

- In Figure 1 we can see to distinct clusters
- Let us standardize the data of Table 1
- The mean age equals m_1 = 37.5 and the mean absolute deviation of the first variable works out to be s_1 = (2.5 + 2.5 + 2.5 + 2.5)/4 = 2.5
- Therefore, standardization converts age 40 to + 1 ((40-37.5)/2.5 = 1)and age 35 ((35 - 37.5)/2.5 = -1) to – 1
- Analogously, m_2 = 175 cm and s_2 = (15 + 15 + 15 + 15)/4 = 15 cm, so 190 cm is standardized to +1 and 160 cm to - 1

Example

- The resulting data matrix, which is unitless, is given in Table 2
- Note that the new averages are zero and that the mean deviations equal 1

- Table 2

Person	Variable 1	Variable 2
A	1	1
B	-1	1
C	1	-1
D	-1	-1

- Even when the data are converted to very strange units standardization will always yield the same numbers

Example

- Plotting the values of Table 2 in Figure 2 does not give a very exciting result
- Figure 2 shows no clustering structure because the four points lie at the vertices of a square
- One could say that there are four clusters, each consisting of a single point, or that there is only one big cluster containing four points
- Here standardizing is no solution

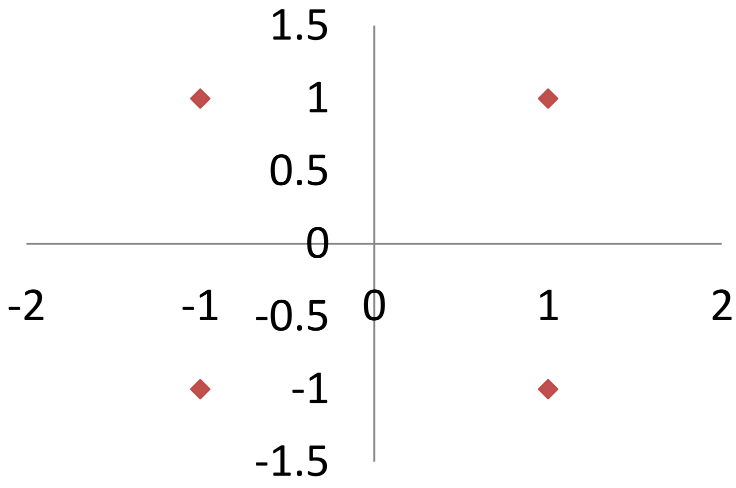


FIGURE: 2

Choice of measurement (Units)- Merits and demerits

- The choice of measurement units gives rise to relative weights of the variables
- Expressing a variable in smaller units will lead to a larger range for that variable, which will then have a large effect on the resulting structure
- On the other hand, by standardizing one attempts to give all variables an equal weight, in the hope of achieving objectivity
- As such, it may be used by a practitioner who possesses no prior knowledge

Choice of measurement- Merits and demerits

- However, it may well be that some variables are intrinsically more important than others in a particular application, and then the assignment of weights should be based on subject-matter knowledge
- On the other hand, there have been attempts to devise clustering techniques that are independent of the scale of the variables

Distances computation between the objects

- The next step is to compute distances between the objects, in order to quantify their degree of dissimilarity
- It is necessary to have a distance for each pair of objects i and j.
- The most popular choice is the Euclidean distance:

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots + (x_{ip} - x_{jp})^2}$$

- When the data are being standardized, one has to replace all x by z in this expression
- This Formula corresponds to the true geometrical distance between the points with coordinates (x_{i1},. .., x_{ip}) and (x_{j1} ,..., x_{jp})

Example

- let us consider the special case with p = 2 (Figure 3)
- Figure shows two points with coordinates (x_{i1} , x_{i2}) and (x_{j1}, x_{j2})
- It is clear that the actual distance between objects i and j is given by the length of the hypotenuse of the triangle, yielding expression in previous slide by virtue of Pythagoras' theorem

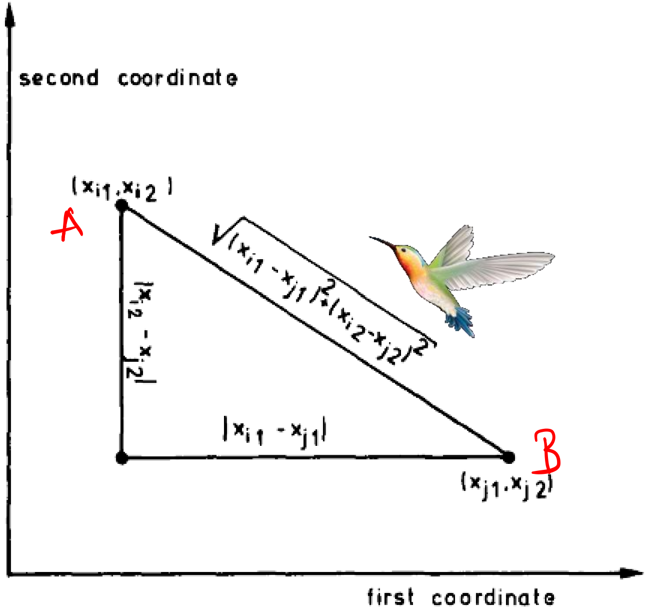
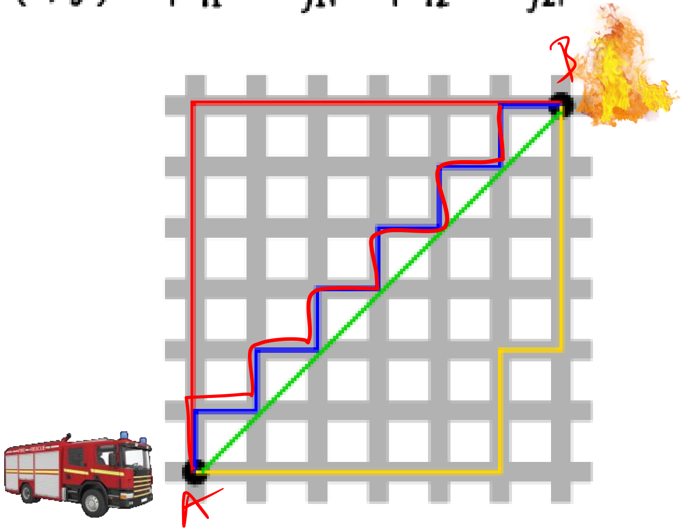


Figure 3: Illustration of the Euclidean distance formula

Distances computation between the objects

- Another well-known metric is the city block or Manhattan distance, defined by:

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots + |x_{ip} - x_{jp}|$$



Interpretation

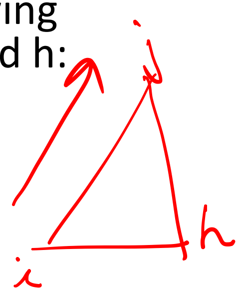
- Suppose you live in a city where the streets are all north-south or east-west, and hence perpendicular to each other
- Let Figure 3 be part of a street map of such a city, where the streets are portrayed as vertical and horizontal lines

Interpretation

- Then the actual distance you would have to travel by car to get from location i to location j would total $|x_{i1} - x_{j1}| + |x_{i2} - x_{j2}|$
- This would be the shortest length among all possible paths from i to j
- Only a bird could fly straight from point i to point j, thereby covering the Euclidean distance between these points

Mathematical Requirements of a Distance Function

- Both the Euclidean metric and the Manhattan metric satisfy the following mathematical requirements of a distance function, for all objects i, j, and h:
- (D1) $d(i, j) \geq 0$
- (D2) $d(i, i) = 0$
- (D3) $d(i, j) = d(j, i)$
- (D4) $d(i, j) \leq d(i, h) + d(h, j)$
- Condition (D1) merely states that distances are nonnegative numbers and (D2) says that the distance of an object to itself is zero
- Axiom (D3) is the symmetry of the distance function
- The triangle inequality (D4) looks a little bit more complicated, but is necessary to allow a geometrical interpretation
- It says essentially that going directly from i to j is shorter than making a detour over object h



Distances computation between the objects

- If $d(i, j) = 0$ does not necessarily imply that $i = j$, because it can very well happen that two different objects have the same measurements for the variables under study
- However, the triangle inequality implies that i and j will then have the same distance to any other object h, because $d(i, h) \leq d(i, j) + d(j, h) = d(j, h)$ and at the same time $d(j, h) \leq d(j, i) + d(i, h) = d(i, h)$, which together imply that $d(i, h) = d(j, h)$

Minkowski distance

- A generalization of both the Euclidean and the Manhattan metric is the Minkowski distance given by:

$$d(i, j) = (|x_{i1} - x_{j1}|^p + |x_{i2} - x_{j2}|^p + \cdots + |x_{in} - x_{jn}|^p)^{1/p},$$

Where p is any real number larger than or equal to 1

- This is also called the Lp metric, with the Euclidean (p = 2) and the Manhattan (p = 1) as special cases

Example for Calculation of Euclidean and Manhattan Distance

- Let x1 = (1, 2) and x2 = (3, 5) represent two objects as in the given Figure The Euclidean distance between the two is $\sqrt{(2^2 + 3^2)} = 3.61$. The Manhattan distance between the two is 2 + 3 = 5.

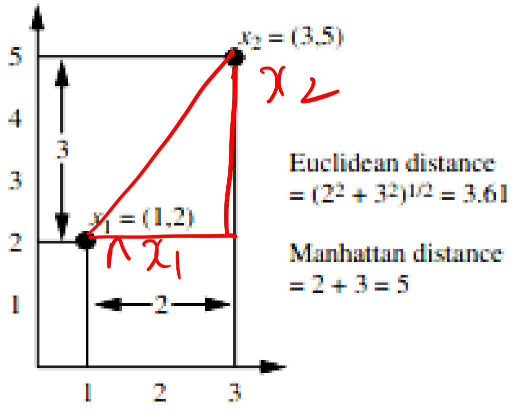


Figure: 4

n- by- n Matrix

- For example, when computing Euclidean distances between the objects of the following Table can be obtain as next slide:

Person	Weight(Kg)	Height(cm)
A	15	95
B	49	156
C	13	95
D	45	160
E	85	178
F	66	176
G	12	90
H	10	78

- Euclidean distances between B and E:
- $((49 - 85)^2 + (156\text{-}178)^2)^{½} = 42.2$

n- by- n Matrix

	A	B	C	D	E	F	G	H
A	0	69.8	2.0	71.6	108.6	95.7	5.8	17.7
B	69.8	0	70.8	5.7	$\boxed{42.2}$	26.3	75.7	87.2
C	2.0	70.8	0	72.5	109.9	96.8	5.1	17.3
D	71.6	5.7	72.5	0	43.9	26.4	77.4	89.2
E	108.6	$\boxed{42.2}$	109.9	43.9	0	19.1	114.3	125.0
F	95.7	26.3	96.8	26.4	19.1	0	101.6	112.9
G	5.8	75.7	5.1	77.4	114.3	101.6	0	12.2
H	17.7	87.2	17.3	89.2	125.0	112.9	12.2	0

Interpretation

- The distance between object B and object E can be located at the intersection of the fifth row and the second column, yielding 42.2
- The same number can also be found at the intersection of the second row and the fifth column, because the distance between B and E is equal to the distance between E and B
- Therefore, a distance matrix is always symmetric
- Moreover, note that the entries on the main diagonal are always zero, because the distance of an object to itself has to be zero

Distance matrix

- It would suffice to write down only the lower triangular half of the distance matrix

	A	B	C	D	E	F	G
B	69.8						
C	2.0	70.8					
D	71.6	5.7	72.5				
E	108.6	42.2	109.9	43.9			
F	95.7	26.3	96.8	26.4	19.1		
G	5.8	75.7	5.1	77.4	114.3	101.6	
H	17.7	87.2	17.3	89.2	125.0	112.9	12.2

Selection of variables

- It should be noted that a variable not containing any relevant information (say, the telephone number of each person) is worse than useless, because it will make the clustering less apparent.
- The Occurrence of several such “trash variables” will kill the whole clustering because they yield a lot of random terms in the distances, thereby hiding the useful information provided by the other variables.
- Therefore, such non informative variables must be given a zero weight in the analysis, which amounts to deleting them

Selection of variables

- The selection of “good” variables is a nontrivial task and may involve quite some trial and error (in addition to subject-matter knowledge and common sense)
- In this respect, cluster analysis may be considered an exploratory technique

Thank you